Practical No 1

Title : Introduction to Android, Introduction to Android Studio IDE, Application
Fundamentals: Creating a Project, Android Components, Activities, Services, Content
Providers, Broadcast Receivers, Interface overview, Creating Android Virtual device, USB
debugging mode, Android Application Overview. Simple "Hello World" program.

Introduction : Android Studio is Google's IDE for Android apps. Android Studio gives you
an advanced code editor and a set of app templates. In addition, it contains tools for development, debugging, testing, and performance that make it faster and easier to develop apps. You can test your apps with a large range of preconfigured emulators or on
your own mobile device, and build production APKs for publication.

Android - Application Components

Application components are the essential building blocks of an Android application. These
components are loosely coupled by the application manifest file *AndroidManifest.xml* that
describes each component of the application and how they interact.
There are following four main components that can be used within an Android application

1. Activity

An *activity* is the entry point for interacting with the user. It represents a single screen with a user interface. In-short Activity performs actions on the screen.

2. Services

A service is a component that runs in the background to perform long-running operations. A service does not provide a user interface. For example, a service might play music in the background while the user is in a different application, or it might fetch data over the network without blocking user interaction with an activity.

3. Broadcast Receivers

They handle communication between Android OS and applications.. Broadcast Receivers simply respond to broadcast messages from other applications or from the system. A *broadcast receiver* is a component that enables the system to deliver events to the app outside of a regular user flow, allowing the app to respond to system-wide broadcast announcements.

4. Content Providers

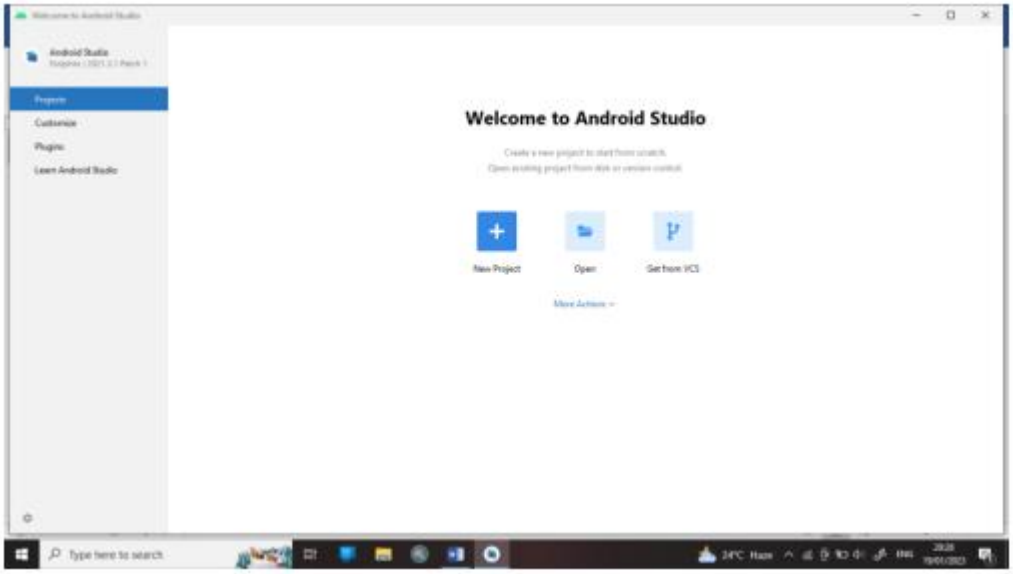
A content provider component supplies data from one application to others on request. A *content provider* manages a shared set of app data that you can store in

the file system, in a SQLite database, on the web, or on any other persistent storage location that your app can access.

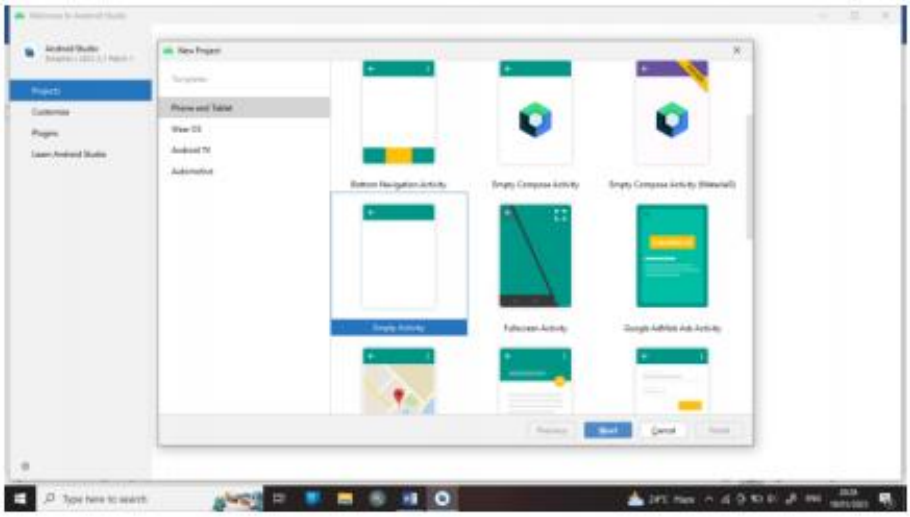
To create a project in Android Studio:

1. Double click the Android Studio icon to launch Android Studio.

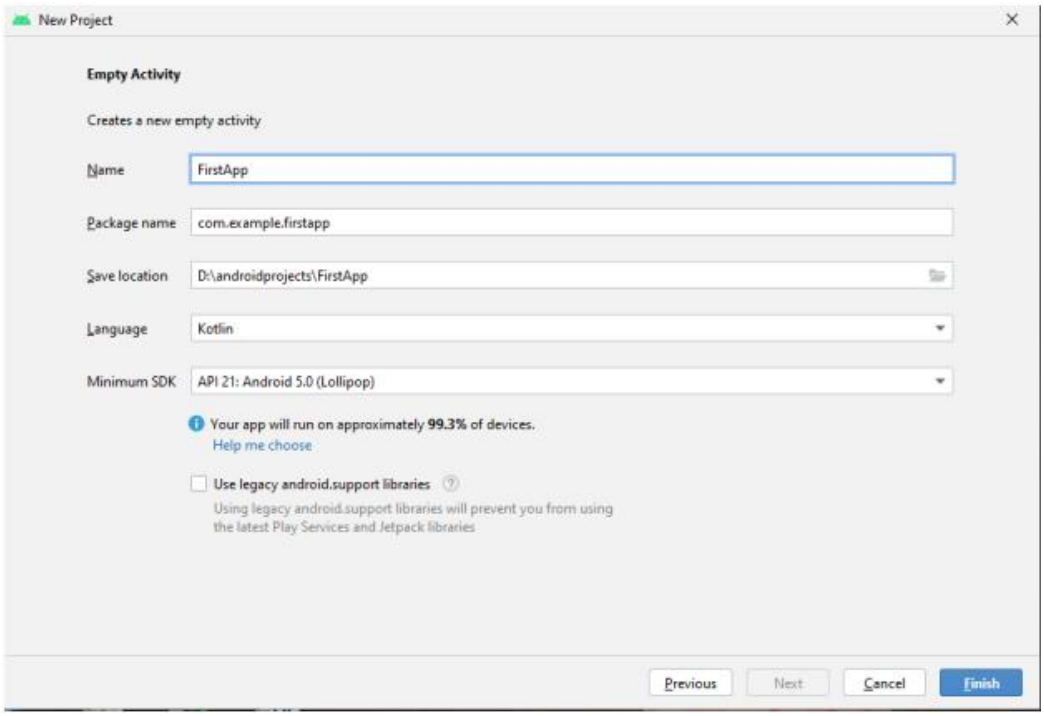
2. In the **Welcome to Android Studio** dialog, click **New Project**.



The **New Project** window opens with a list of templates provided by Android Studio. In Android Studio, a project template is an Android project that provides the blueprint for a
certain type of app. Templates create the structure of the project and the files needed for
Android Studio to build your project. Based on the template that you choose, it provides
starter code to get you going faster.

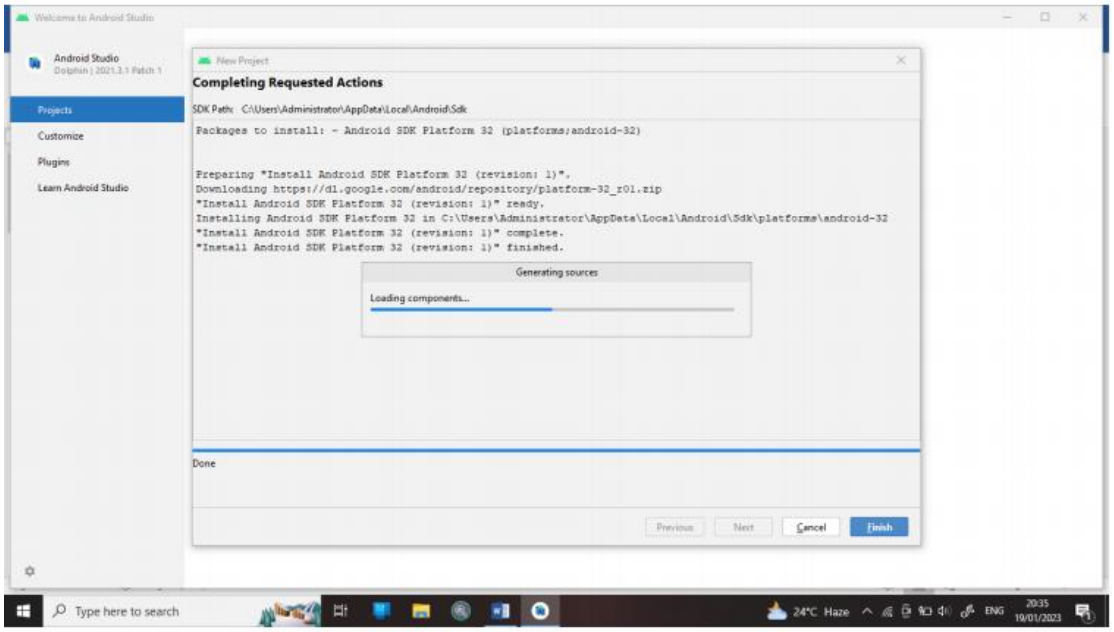


3. Make sure the **Phone and Tablet** tab is selected.
4. Click the **Empty Compose Activity** template to select it as the template for your project.
The **Empty Compose Activity template** is the template to create a simple project that you can
use to build a Compose app. It has a single screen and displays the text "Hello World !".
5. Click **Next**. The **New Project** dialog opens. This has some fields to configure your project.

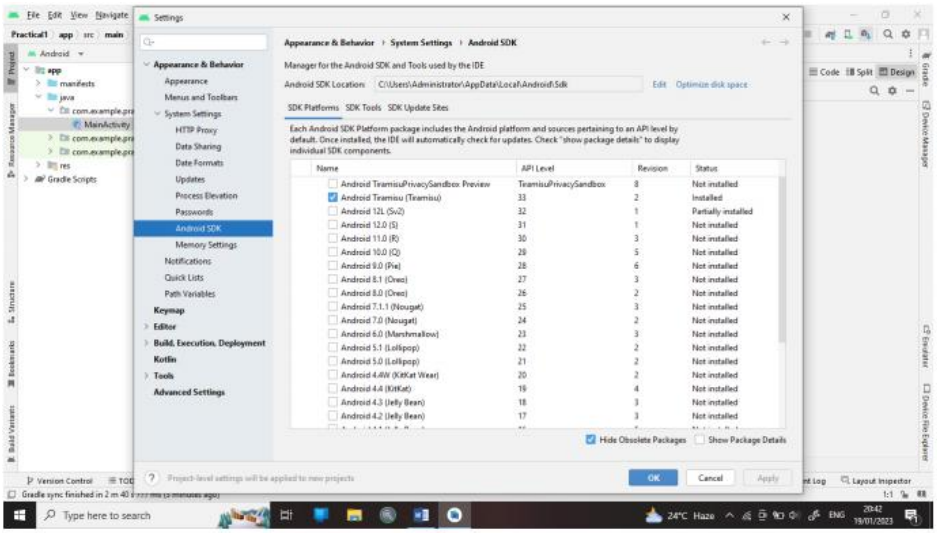


The **Name** field is used to enter the name of your project, for this codelab type "FirstApp".
Leave the **Package name** field as is. This is how your files will be organized in the file structure.
In this case, the package name will be com.example.FirstApp
Select the **Save location** field as you required. It contains the location where all the files
related to your project are saved.

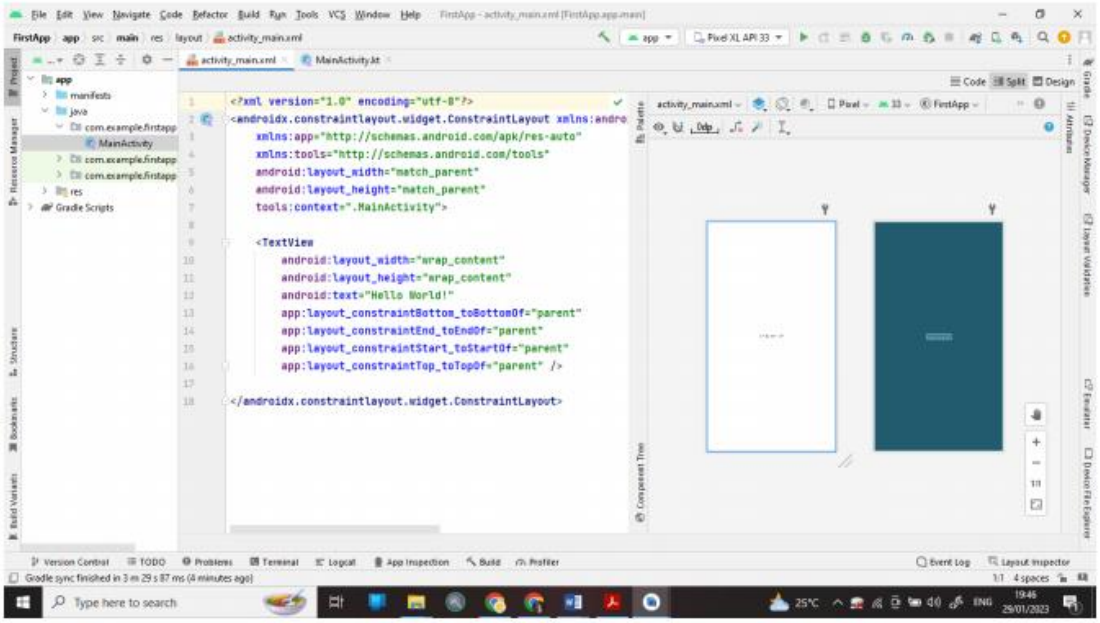
Kotlin is already selected in the **Language** field. Language defines which programming
language you want to use for your project. Since Compose is only compatible with Kotlin, you
cannot alter this field.
Select **API 21: Android 5.0 (Lollipop)** from the menu in the **Minimum SDK** field. Minimum
SDK indicates the minimum version of Android that your app can run on.
The **Use legacy android.support libraries** checkbox is already unchecked.
6. Hit **Finish**. This may take a while depending upon speed of your Internet Connection .While
Android Studio is setting up, a progress bar and message indicates whether Android Studio is
still setting up your project.



Downloads required SDK components while setup. In Android Studio, select Tools > SDK Manager, or click the SDK Manager icon in the toolbar to check installed SDK in system.



Click **Split** on the top right of Android Studio, this allows you to view both code and design. You can also click **Code** to view code only or click **Design** to view design only.



Create a virtual device (Emulator)

Device Manager is used to create a virtual device or emulator that simulates the configuration for a particular type of Android device. Using the Device Manager, define the hardware characteristics of a device and its API level, and save it as a virtual device configuration. When you start the Android emulator, it reads a specified configuration and creates an emulated device that behaves exactly like a physical version of that device , but it resides on your computer .

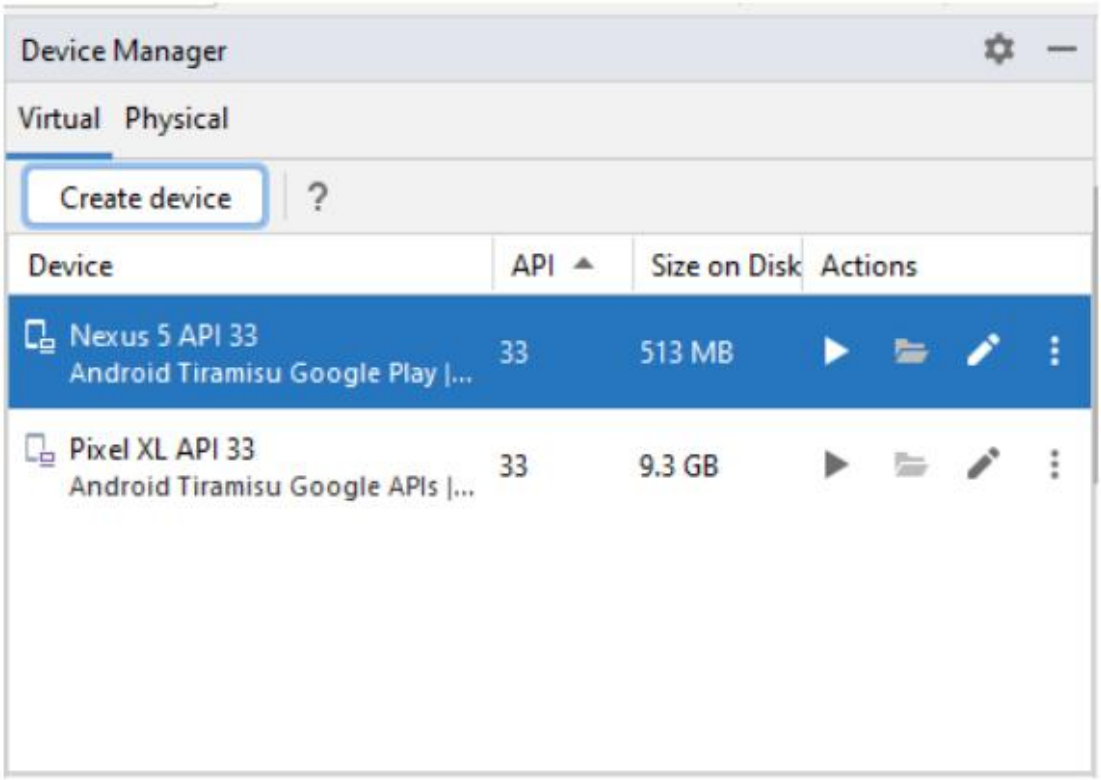
Create a virtual device

In Android Studio, select Tools > Device Manager, or click the Device Manager icon in the toolbar.

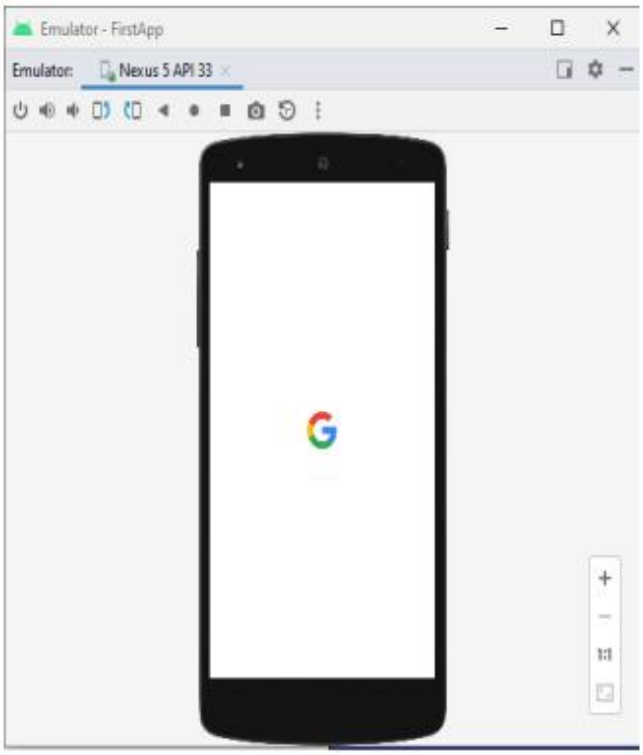
1. Click the **Create Device**…. (If you have created a virtual device before, the window shows
all of your existing devices and the button is at the bottom.)

Run your app on an emulator

1. In Android Studio, select **Run > Run app** or click the **Run icon**
in the toolbar.
2. In the **Device Manager** window, under **Available Emulators**, select **Nexus 5 API 33** and
click **OK**.



The emulator starts and boots just like a physical device. Depending on the speed of your computer, this may take a while. Your app builds, and once the emulator is ready, Android Studio will upload the app to the emulator and run it.



Run your app on a device

Run your app on a physical mobile device such as a phone or tablet.

Requirement:

An Android device such as a phone or tablet. A data cable to connect your Android device to your computer via the USB port. To let Android Studio communicate with your device, turn on USB Debugging on your Android device. This is enabled in the Developer options settings of your device. Note this is not the same as rooting your device. On Android 4.2 and higher, the Developer options screen is hidden by default. To show Developer options and enable USB Debugging:

1. On your device, open **Settings > About phone** and tap **Build number** seven times.
2. Return to the previous screen (**Settings**). **Developer options** appears at the bottom of the
list. Click **Developer options**.
3. Choose **USB Debugging**.

Practical No 2

Title : Programming Resources
Android Resources: (Color, Theme, String, Drawable, Dimension, Image),

Color.xml

```
<color name="blue">#2196F3</color>
```

String.xml

```
<resources>
    <string name="app_name">Programming Resources</string>
    <string name="title_text">Android Resource Demo</string>
</resources>
```

Bg_shape.xml

```
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#4CAF50"/>
    <corners android:radius="12dp"/>
</shape>
```

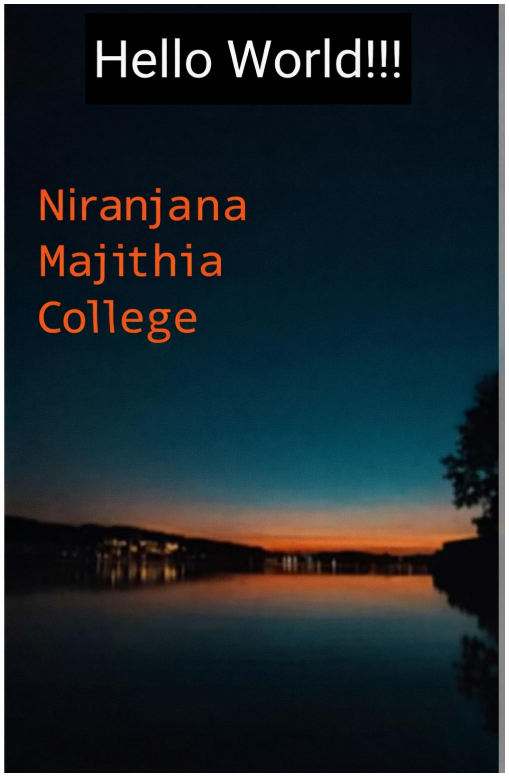
Theme.xml

```
<style name="Theme.MyApp" parent="Theme.Material3.DayNight.NoActionBar">
    <item name="colorPrimary">@color/primaryColor</item>
</style>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="@dimen/padding_size"
    android:background="@color/primaryColor">

    <TextView
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="@string/title_text"
        android:textColor="@color/textColor"
        android:textSize="@dimen/text_size"/>

    <ImageView
        android:layout_width="150dp"
        android:layout_height="150dp"
        android:layout_marginTop="20dp"
        android:src="@drawable/logo"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Resource Button"
        android:background="@drawable/bg_shape"
        android:textColor="@color/textColor"/>

</LinearLayout>
```



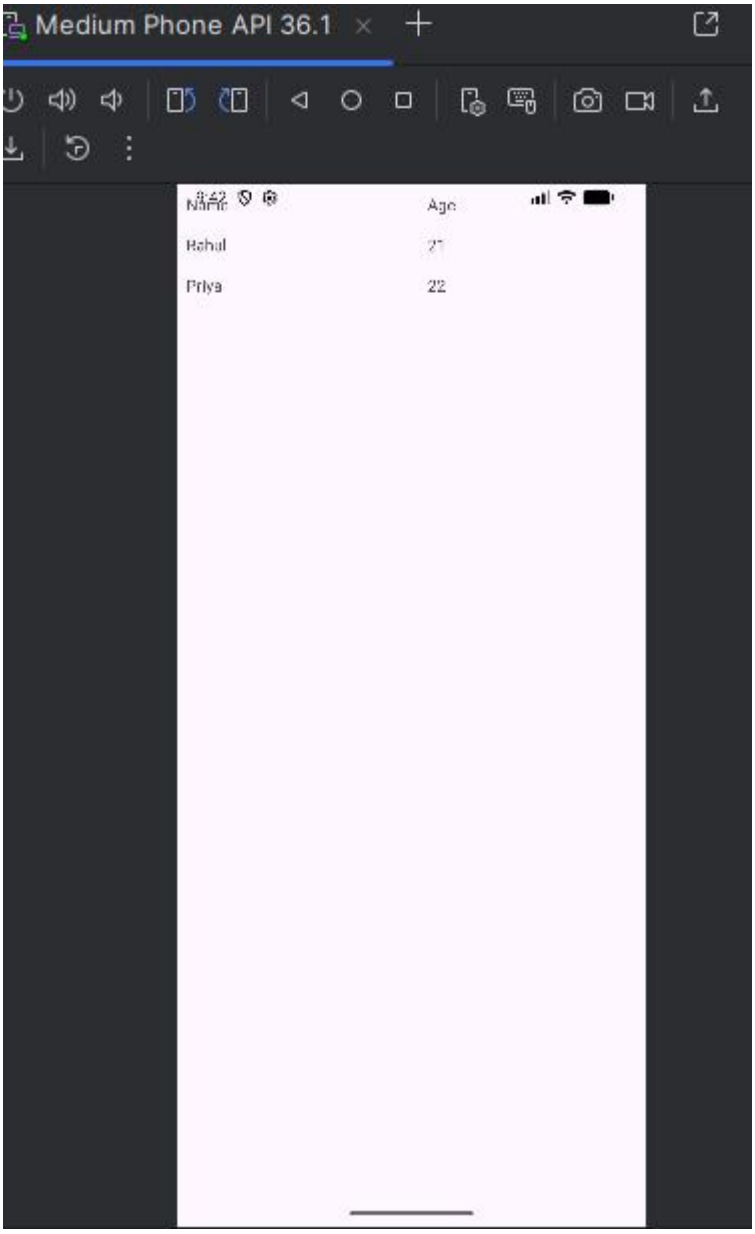
Practical 3

MainAcitivity .kt

```
package com.example.tablelayout

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import com.example.myapplication.R

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:stretchColumns="*">

    <TableRow>
        <TextView
            android:text="Name"
            android:padding="8dp"/>

        <TextView
            android:text="Age"
            android:padding="8dp"/>
    </TableRow>

    <TableRow>
        <TextView
            android:text="Rahul"
            android:padding="8dp"/>

        <TextView
            android:text="21"
            android:padding="8dp"/>
    </TableRow>

    <TableRow>
        <TextView
            android:text="Priya"
            android:padding="8dp"/>

        <TextView
            android:text="22"
            android:padding="8dp"/>
    </TableRow>

</TableLayout>
```

output



Relative layout

```
package com.example.relativelayout

import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import com.example.myapplication.R

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        val name = findViewById<EditText>(R.id.nameInput)
        val age = findViewById<EditText>(R.id.ageInput)
        val button = findViewById<Button>(R.id.submitBtn)

        button.setOnClickListener {
            Toast.makeText(
                this,
                "Name: ${name.text}    Age: ${age.text}",
                Toast.LENGTH_SHORT
            ).show()
        }
    }
}
```

activity.main

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <!-- Title -->
    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="User Details"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_centerHorizontal="true"/>

    <!-- Name -->
    <TextView
        android:id="@+id/nameLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name:"
        android:layout_below="@id/title"
        android:layout_marginTop="30dp"/>

    <EditText
        android:id="@+id/nameInput"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:layout_toEndOf="@id/nameLabel"
        android:hint="Enter name"/>

    <!-- Age -->
    <TextView
        android:id="@+id/ageLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Age:"
        android:layout_below="@id/nameLabel"
        android:layout_marginTop="20dp"/>
```

```
    <EditText
        android:id="@+id/ageInput"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:layout_toEndOf="@id/ageLabel"
        android:hint="Enter age"/>

    <!-- Button -->
    <Button
        android:id="@+id/submitBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/ageInput"
        android:layout_marginTop="125dp"
        android:text="Submit" />

</RelativeLayout>
```

manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplication">
        <activity
          android:name="com.example.relativelayout.MainActivity"
            android:exported="true"
            android:theme="@style/Theme.MyApplication">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Linear layout

Mainactivity.kt

```
package com.example.myapplication

import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val start = findViewById<Button>(R.id.btn1)
        val pause = findViewById<Button>(R.id.btn2)
        val stop = findViewById<Button>(R.id.btn3)

        start.setOnClickListener {
            Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show()
        }

        pause.setOnClickListener {
            Toast.makeText(this, "Service Paused", Toast.LENGTH_SHORT).show()
        }

        stop.setOnClickListener {
            Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show()
        }
    }
}
```

activity.main.xml

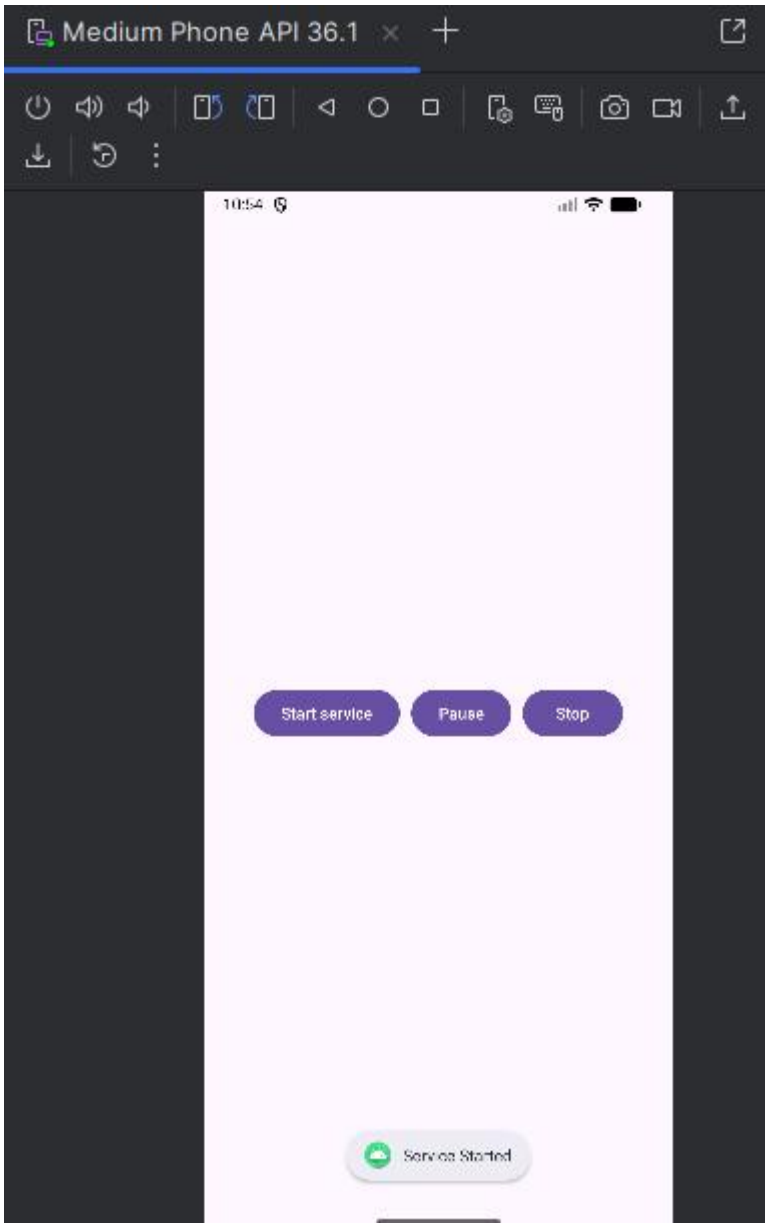
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal"
    android:gravity="center"
    android:padding="16dp">

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start service"
        android:layout_marginEnd="10dp"/>

    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Pause"
        android:layout_marginEnd="10dp"/>

    <Button
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Stop"/>

</LinearLayout>
```

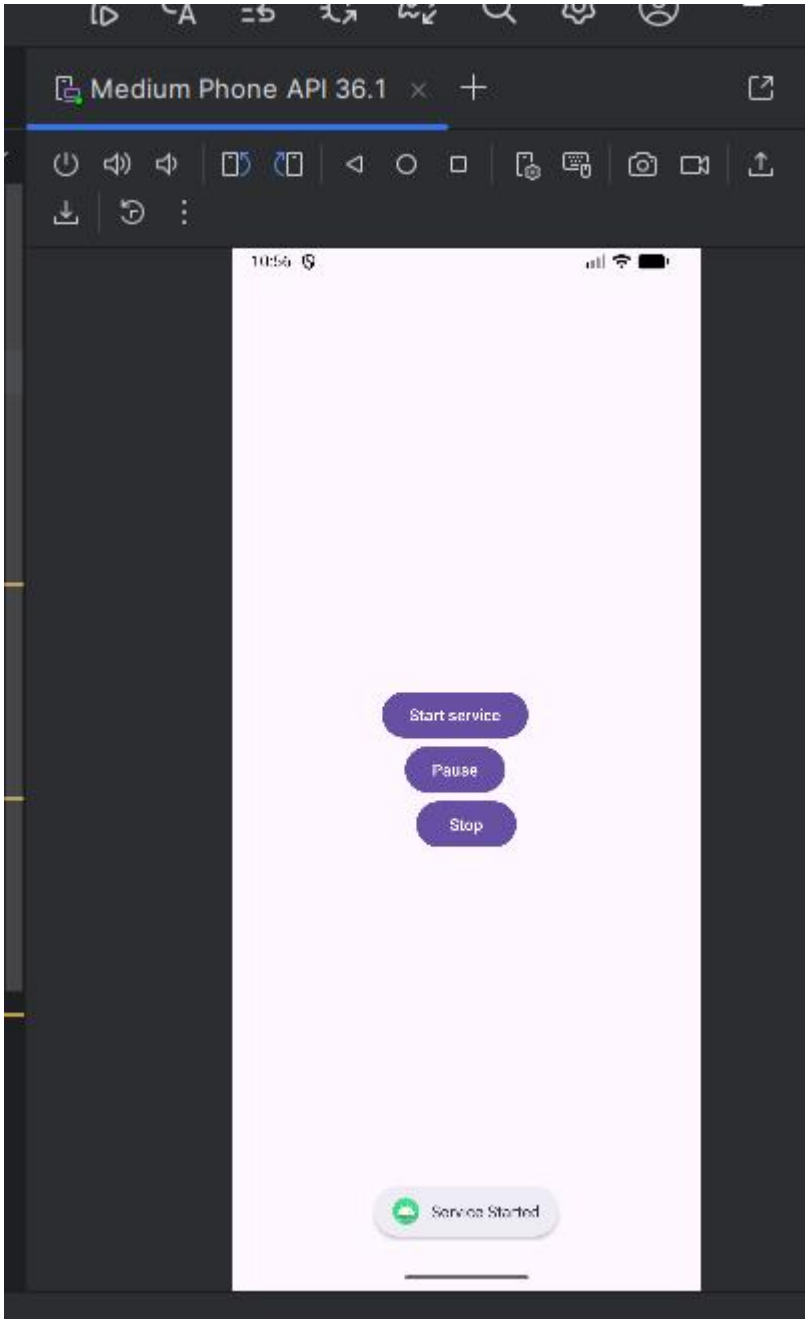
Colors.xml

```
<resources>
    <color name="teal_700">#008577</color>
    <color name="black">#000000</color>
</resources>
```

Dimen.xml

```
<resources>
    <dimen name="btnwidth">200dp</dimen>
</resources>
```





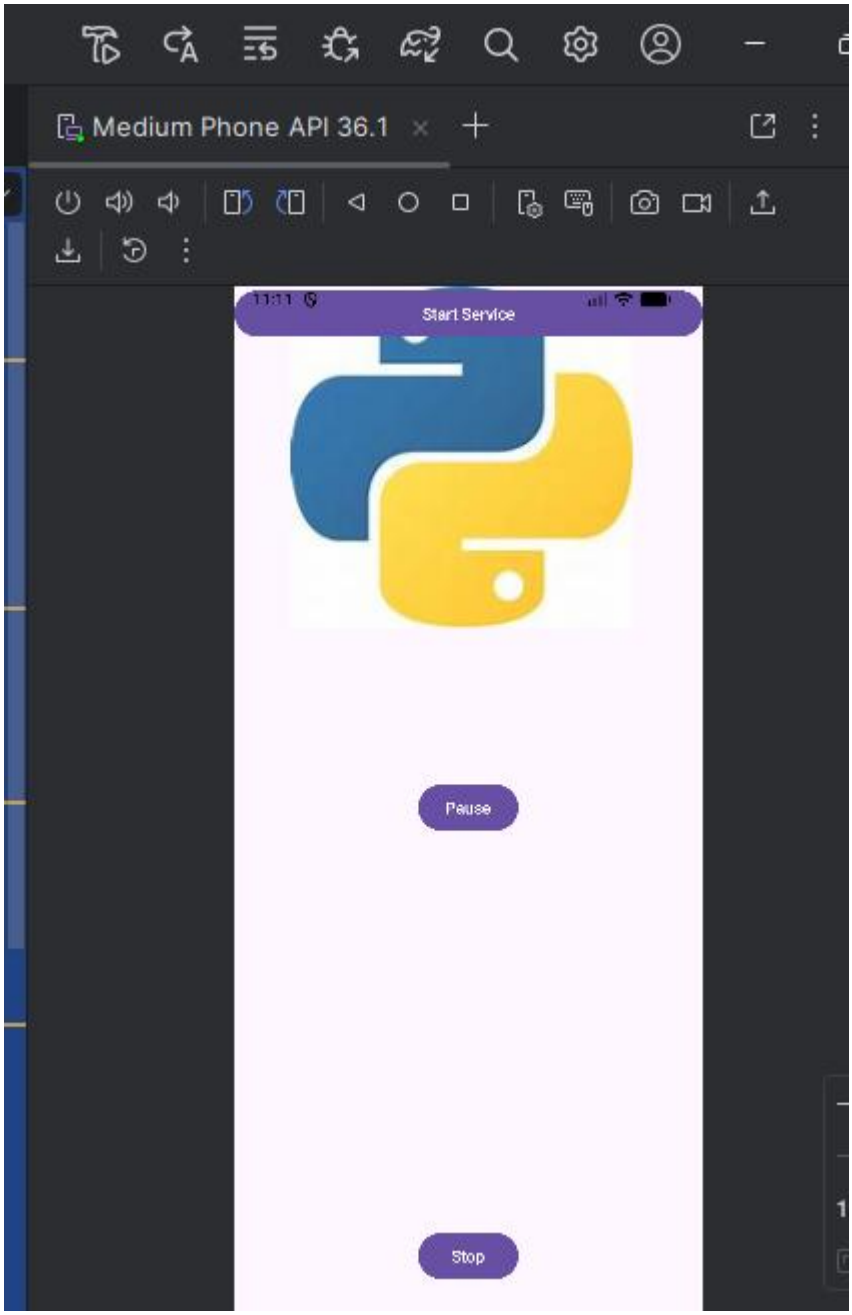
Frame layout

```
package com.example.myapplication

import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        val start = findViewById<Button>(R.id.btn1)
        val pause = findViewById<Button>(R.id.btn2)
        val stop = findViewById<Button>(R.id.btn3)

        start.setOnClickListener {
            Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show()
        }

        pause.setOnClickListener {
            Toast.makeText(this, "Service Paused", Toast.LENGTH_SHORT).show()
        }

        stop.setOnClickListener {
            Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show()
        }
    }
}
```

activity .main xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <ImageView
        android:layout_width="400dp"
        android:layout_height="300dp"
        android:src="@drawable/python" />
    <!-- Background Button -->
    <Button
        android:id="@+id/btn1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start Service" />

    <!-- Center Button (on top) -->
    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Pause"
        android:layout_gravity="center" />

    <!-- Bottom Button (topmost layer) -->
    <Button
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Stop"
        android:layout_gravity="bottom|center_horizontal"
        android:layout_marginBottom="40dp"/>

</FrameLayout>
```



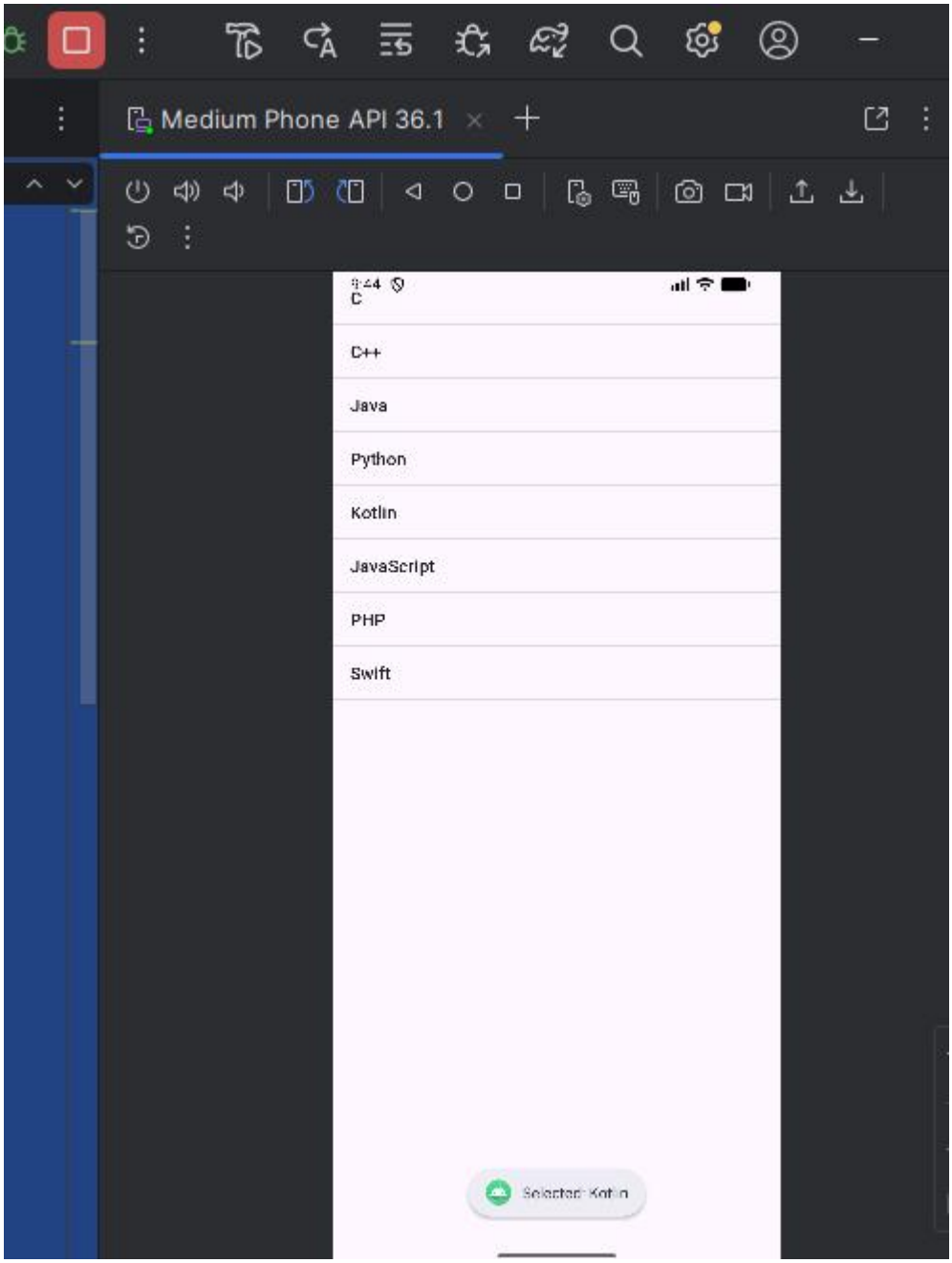
ListView
Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ListView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/listViewLanguages"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

Pract4.kt

```
package com.example.pract4

import android.os.Bundle
import android.widget.ArrayAdapter
import android.widget.ListView
```

```
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import com.example.pract4.R

class MainActivity : AppCompatActivity() {

    private lateinit var listView: ListView

    private val languages = arrayOf(
        "C",
        "C++",
        "Java",
        "Python",
        "Kotlin",
        "JavaScript",
        "PHP",
        "Swift"
    )

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        listView = findViewById(R.id.listViewLanguages)

        val adapter = ArrayAdapter(
            this,
            android.R.layout.simple_list_item_1,        language )

        listView.adapter = adapter

        listView.setOnItemClickListener { _, _, position, _ ->
            Toast.makeText(
                this,
                "Selected: ${languages[position]}",
                Toast.LENGTH_SHORT
            ).show()
        }
    }
}
```

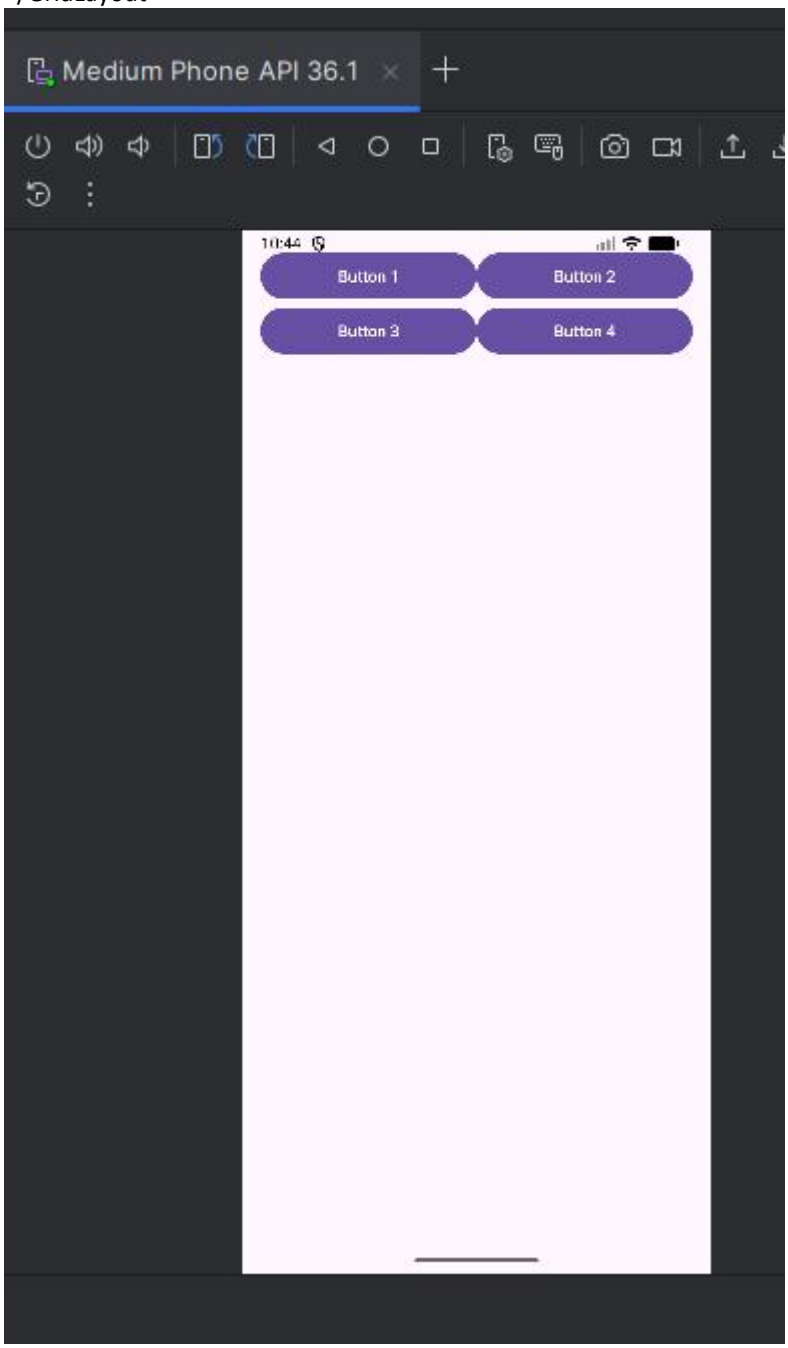


Gridlayout
Gridlayoutdemo.kt

```
package com.example.gridlayoutdemo

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.Toast

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        val btn1 = findViewById<Button>(R.id.btn1)
        val btn2 = findViewById<Button>(R.id.btn2)
        val btn3 = findViewById<Button>(R.id.btn3)
        val btn4 = findViewById<Button>(R.id.btn4)

        btn1.setOnClickListener {
            Toast.makeText(this, "Button 1 Clicked", Toast.LENGTH_SHORT).show()
        }

        btn2.setOnClickListener {
            Toast.makeText(this, "Button 2 Clicked", Toast.LENGTH_SHORT).show()
        }

        btn3.setOnClickListener {
            Toast.makeText(this, "Button 3 Clicked", Toast.LENGTH_SHORT).show()
        }

        btn4.setOnClickListener {
            Toast.makeText(this, "Button 4 Clicked", Toast.LENGTH_SHORT).show()
        }
    }
}

<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:columnCount="2"
    android:rowCount="2"
    android:padding="16dp">

    <Button
        android:id="@+id/btn1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="Button 1" />

    <Button
        android:id="@+id/btn2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="Button 2" />

    <Button
        android:id="@+id/btn3"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="Button 3" />

    <Button
        android:id="@+id/btn4"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
```

```
        android:layout_columnWeight="1"
        android:text="Button 4" />

</GridLayout>
```



Practical 6

a) Pop-up Menu
b) Context Menu
c) Option Menu

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnPopup"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show Popup Menu"/>

    <TextView
        android:id="@+id/txtContext"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Long press me for Context Menu"
        android:textSize="18sp"
        android:padding="15dp"
        android:background="#DDDDDD"
        android:layout_marginTop="20dp"
        android:gravity="center"/>

</LinearLayout>
```

popup_menu.xml

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:id="@+id/popup_edit" android:title="Edit"/>
    <item android:id="@+id/popup_delete" android:title="Delete"/>
</menu>
```

context_menu.xml

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:id="@+id/context_copy" android:title="Copy"/>
    <item android:id="@+id/context_paste" android:title="Paste"/>
</menu>
```

option_menu.xml

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:id="@+id/opt_settings" android:title="Settings"/>
    <item android:id="@+id/opt_exit" android:title="Exit"/>
</menu>
```

MainActivity.kt

```
package com.example.pract8
import android.os.Bundle
import android.view.*
import android.widget.Button
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.appcompat.widget.PopupMenu

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
```

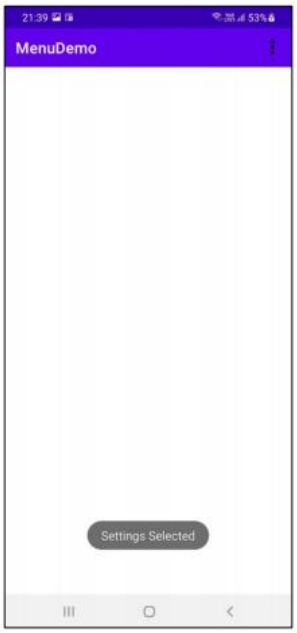
```
    setContentView(R.layout.activity_main)

    val btnPopup = findViewById<Button>(R.id.btnPopup)
    val txtContext = findViewById<TextView>(R.id.txtContext)

    // Register Context Menu
    registerForContextMenu(txtContext)

    // Popup Menu
    btnPopup.setOnClickListener {
        val popup = PopupMenu(this, btnPopup)
        popup.menuInflater.inflate(R.menu.popup_menu, popup.menu)

        popup.setOnMenuItemClickListener {
            Toast.makeText(this, it.title, Toast.LENGTH_SHORT).show()
            true
        }
        popup.show()
    }
}

// Context Menu
override fun onCreateContextMenu(
    menu: ContextMenu?,
    v: View?,
    menuInfo: ContextMenu.ContextMenuInfo?
) {
    super.onCreateContextMenu(menu, v, menuInfo)
    menuInflater.inflate(R.menu.context_menu, menu)
}
```

```
    override fun onContextItemSelected(item: MenuItem): Boolean {
        Toast.makeText(this, item.title, Toast.LENGTH_SHORT).show()
        return true
    }

    // Option Menu
    override fun onCreateOptionsMenu(menu: Menu?): Boolean {
        menuInflater.inflate(R.menu.option_menu, menu)
        return true
    }

    override fun onOptionsItemSelected(item: MenuItem): Boolean {
        Toast.makeText(this, item.title, Toast.LENGTH_SHORT).show()
        return true
    }
}
```

MenuDemo
CLICK ME
You Clicked Kotlin

MenuDemo
Long press on List Item to Open Context Menu
Siddhi
Sneha
Manasi
Neha
Pooja
Shriya
Maya
Riya
Khushi
Sanika
Samata
Isha
Aparna
WhatsApp Manasi
Sonal

MenuDemo
Settings
Share
Exit


MenuDemo
Settings Selected

Practical 7

MainActivity.kt

```
package com.example.myapplication


import android.content.Intent
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val editName = findViewById<EditText>(R.id.editName)
        val btnSend = findViewById<Button>(R.id.btnSend)

        btnSend.setOnClickListener {

            val name = editName.text.toString()

            val intent = Intent(this, SecondActivity::class.java)
            intent.putExtra("USERNAME", name)

            startActivity(intent)
        }
    }
}
```

SecondActivity.kt

```
package com.example.myapplication

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
```

```
class SecondActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_second)

        val txtResult = findViewById<TextView>(R.id.txtResult)

        val receivedName = intent.getStringExtra("USERNAME")

        txtResult.text = "Welcome $receivedName"
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplication">

        <!-- Main Activity -->
        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>

        <!-- Second Activity -->
        <activity
            android:name=".SecondActivity"
            android:exported="false"/>
```

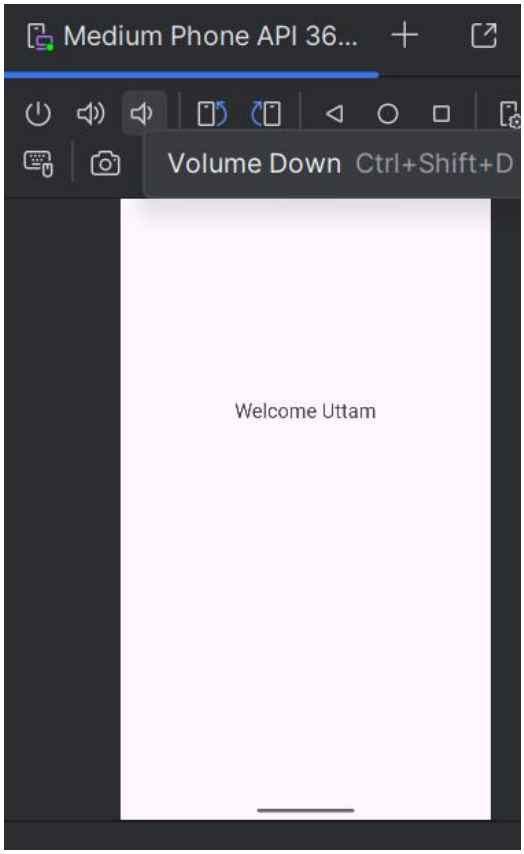
```
    </application>

</manifest>
```

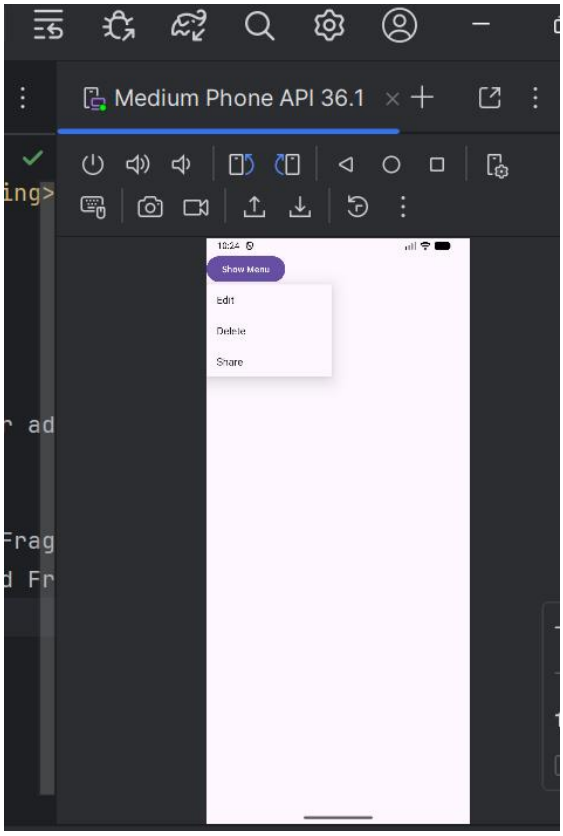
Activity_main.xml

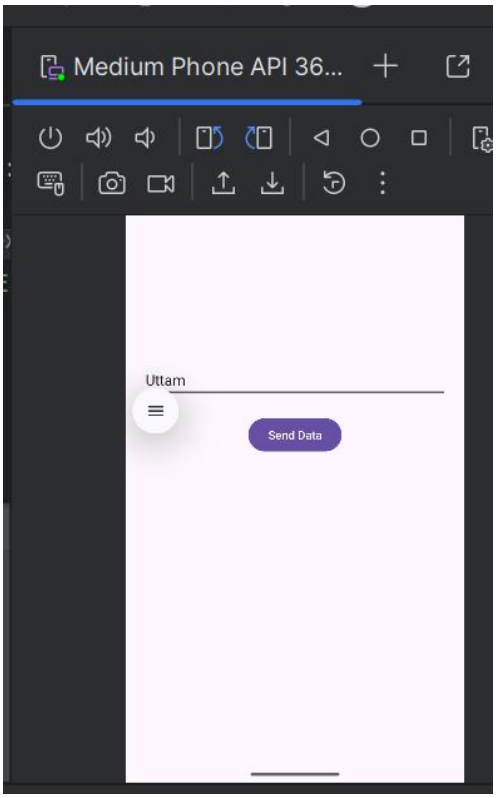
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/editName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Your Name"/>

    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Data"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

Activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:id="@+id/txtResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="22sp"/>

</LinearLayout>
```


Medium Phone API 36...
Volume Down Ctrl+Shift+D
Welcome Uttam


Medium Phone API 36.1
Show Menu
Edit
Delete
Share


Medium Phone API 36...
Uttam
Send Data

Practical 8

MainActivity.kt

```
package com.example.pract8

import android.app.Service
import android.content.Intent
import android.media.MediaPlayer
import android.os.IBinder
import android.widget.Toast

class NewService : Service() {

    private lateinit var player: MediaPlayer

    override fun onCreate() {
        super.onCreate()
        player = MediaPlayer.create(this, R.raw.music)
        player.isLooping = true
        Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show()
    }

    override fun onStartCommand(intent: Intent?, flags: Int, startId: Int): Int {
        player.start()
        return START_STICKY
    }

    override fun onDestroy() {
        super.onDestroy()
        player.stop()
        Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show()
    }

    override fun onBind(intent: Intent?): IBinder? {
        return null
    }
}
```

NewService.kt

```
package com.example.servicesdemo
import android.app.Service
import android.content.Intent
import android.media.MediaPlayer
import android.os.IBinder
import android.provider.Settings
class NewService : Service() {
   // declaring object of MediaPlayer
   private lateinit var player: MediaPlayer
   // execution of service will start on calling this method
   override fun onStartCommand(intent: Intent, flags: Int, startId: Int): Int {
      // creating a media player which will play the audio of Default ringtone in android device
      player = MediaPlayer.create(this,Settings.System.DEFAULT_RINGTONE_URI)
      // providing the boolean value as true to play the audio on loop
      player.setLooping(true)
      //starting the process
      player.start()
      // returns the status of the program
      return START_STICKY
   }
   // execution of the service will stop on calling this method
   override fun onDestroy() {
      super.onDestroy()
      //stopping the process
      player.stop()
   }
   override fun onBind(intent: Intent?): IBinder? {
      return null
   }
}
```

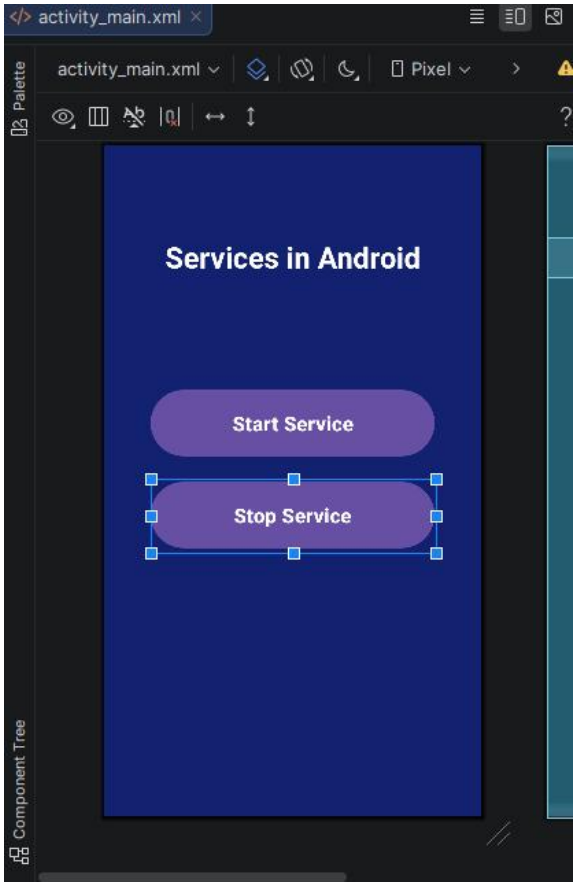
Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:background="#132171"
   tools:context=".MainActivity">
```

```
    <TextView
        android:id="@+id/txtheading"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Services in Android"
        android:gravity="center"
        android:textColor="@android:color/white"
        android:textSize="32sp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/btnstart"
        android:layout_width="310dp"
        android:layout_height="80dp"
        android:layout_gravity="center"
        android:layout_marginTop="120dp"
        android:text="Start Service"
        android:textSize="22sp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/btnstop"
        android:layout_width="310dp"
        android:layout_height="80dp"
        android:layout_gravity="center"
        android:layout_marginTop="20dp"
        android:text="Stop Service"
        android:textSize="22sp"
        android:textStyle="bold" />

</LinearLayout>
```

AndroidMenifest.xml

```
    <service android:name=".NewService" />
    </application>

</manifest>
```



Activity-main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnNonClickable"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show Non-Clickable Notification" />

    <Button
        android:id="@+id/btnClickable"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Show Clickable Notification"
        android:layout_marginTop="20dp"/>
```

```
</LinearLayout>
```

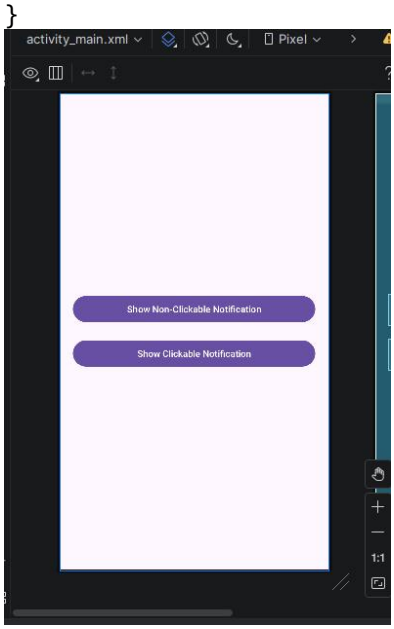
AndroidMenifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools">
   <uses-permission android:name="android.permission.POST_NOTIFICATIONS"/>
```

MainActivity.kt

```
package com.example.pract8
import android.app.*
import android.content.Intent
import android.os.Build
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.NotificationCompat
import androidx.core.app.NotificationManagerCompat

class MainActivity : AppCompatActivity() {

    private val channelId = "my_channel"

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        createNotificationChannel()

        findViewById<Button>(R.id.btnNonClickable).setOnClickListener {
            showNonClickableNotification()
        }

        findViewById<Button>(R.id.btnClickable).setOnClickListener {
            showClickableNotification()
        }
    }

    private fun showNonClickableNotification() {
        val notification = NotificationCompat.Builder(this, channelId)
            .setSmallIcon(android.R.drawable.ic_dialog_info)
            .setContentTitle("Non-Clickable Notification")
            .setContentText("This notification cannot be clicked")
            .setAutoCancel(false)
            .build()

        NotificationManagerCompat.from(this).notify(1, notification)
    }

    private fun showClickableNotification() {
```

```
        val intent = Intent(this, MainActivity::class.java)
        intent.flags = Intent.FLAG_ACTIVITY_NEW_TASK or
Intent.FLAG_ACTIVITY_CLEAR_TASK

        val pendingIntent = PendingIntent.getActivity(
            this, 0, intent, PendingIntent.FLAG_UPDATE_CURRENT or
PendingIntent.FLAG_IMMUTABLE
        )

        val notification = NotificationCompat.Builder(this, channelId)
            .setSmallIcon(android.R.drawable.ic_dialog_info)
            .setContentTitle("Clickable Notification")
            .setContentText("Tap to reload app")
            .setContentIntent(pendingIntent)
            .setAutoCancel(true)
            .build()

        NotificationManagerCompat.from(this).notify(2, notification)
    }

    private fun createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            val channel = NotificationChannel(
                channelId,
                "My Channel",
                NotificationManager.IMPORTANCE_DEFAULT
            )
            val manager = getSystemService(NotificationManager::class.java)
            manager.createNotificationChannel(channel)
        }
    }
}
```



Create an android application which automatically notify the user when Aeroplane mode is turned on or off using broadcast receiver.

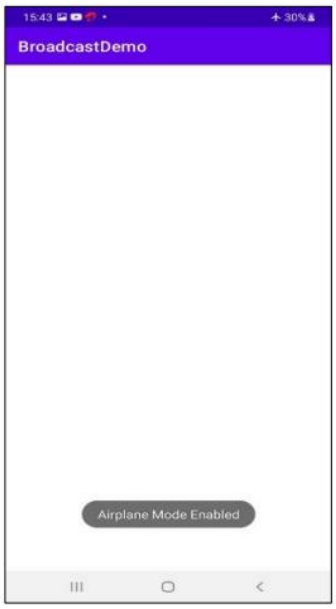
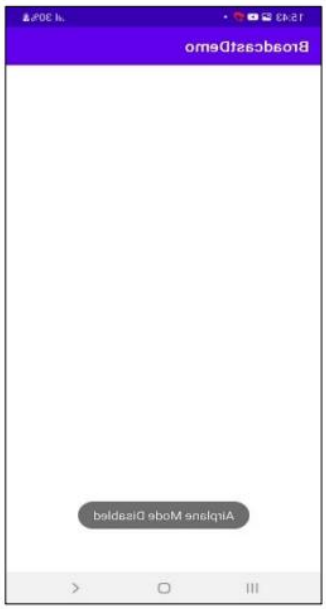
AirplaneReceiver.kt

```
package com.example.pract8

import android.content.BroadcastReceiver
import android.content.Context
import android.content.Intent
import android.provider.Settings
import android.widget.Toast

class AirplaneReceiver : BroadcastReceiver() {

    override fun onReceive(context: Context, intent: Intent) {
        val isOn = Settings.Global.getInt(
            context.contentResolver,
            Settings.Global.AIRPLANE_MODE_ON, 0
        ) != 0

        if (isOn) {
            Toast.makeText(context, "Airplane Mode ON", Toast.LENGTH_LONG).show()
        } else {
            Toast.makeText(context, "Airplane Mode OFF", Toast.LENGTH_LONG).show()
        }
    }
}
```

AndroidMenifest.xml

```
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>
```

```
<receiver android:name=".AirplaneReceiver">
    <intent-filter>
        <action android:name="android.intent.action.AIRPLANE_MODE"/>
    </intent-filter>
</receiver>
```

MainActivity.kt

```
package com.example.pract8

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }
}
```